

Python Programs - Code & Sample Output

4 Programs

1. Simple Calculator

Program Code

```
from tkinter import *

def calculate(op):
    try:
        num1 = float(entry1.get())
        num2 = float(entry2.get())

        if op == "+":
            result = num1 + num2
        elif op == "-":
            result = num1 - num2
        elif op == "*":
            result = num1 * num2
        elif op == "/":
            if num2 != 0:
                result = num1 / num2
            else:
                result = "Cannot divide by zero"

        lbl_result.config(text="Result = " + str(result))

    except ValueError:
        lbl_result.config(text="Please enter valid numbers")

root = Tk()
root.title("Simple Calculator")
root.geometry("300x250")

Label(root, text="First Number").pack()
entry1 = Entry(root)
entry1.pack()

Label(root, text="Second Number").pack()
entry2 = Entry(root)
entry2.pack()

Button(root, text="ADD", width=10,
        command=lambda: calculate("+")).pack()

Button(root, text="SUBTRACT", width=10,
        command=lambda: calculate("-")).pack()

Button(root, text="MULTIPLY", width=10,
        command=lambda: calculate("*")).pack()

Button(root, text="DIVIDE", width=10,
        command=lambda: calculate("/")).pack()

lbl_result = Label(root, text="Result =")
lbl_result.pack()

root.mainloop()
```

Sample Output

Simple Calculator

First Number: 10
Second Number: 5

[ADD]
[SUBTRACT]
[MULTIPLY]
[DIVIDE]

Result = 15.0

2. Tkinter Calculator

Program Code

```
import tkinter as tk

def calculate():
    try:
        a = float(num1.get())
        b = float(num2.get())
        op = operation.get()

        if op == "+":
            result = a + b
        elif op == "-":
            result = a - b
        elif op == "*":
            result = a * b
        elif op == "/":
            result = a / b

        output.config(text="Result: " + str(result))

    except:
        output.config(text="Invalid Input")

root = tk.Tk()
root.title("Calculator")
root.geometry("350x300")

tk.Label(root, text="Number 1").pack()
num1 = tk.Entry(root)
num1.pack()

tk.Label(root, text="Number 2").pack()
num2 = tk.Entry(root)
num2.pack()

operation = tk.StringVar()
operation.set("+")

tk.OptionMenu(root, operation, "+", "-", "*", "/").pack()

tk.Button(root, text="Calculate", command=calculate).pack(pady=10)

output = tk.Label(root, text="Result:")
output.pack()

root.mainloop()
```

Sample Output

```
Calculator

Number 1: 20
Number 2: 5
Operation: +

[ Calculate ]

Result: 25.0
```

3. Student Grade Calculator

Program Code

```
import tkinter as tk

def calculate():
    marks = [
        int(m1.get()),
        int(m2.get()),
        int(m3.get())
    ]

    total = sum(marks)
    average = total / 3

    if average >= 90:
        grade = "A"
    elif average >= 75:
        grade = "B"
    elif average >= 50:
        grade = "C"
    else:
        grade = "Fail"

    result.config(
        text=f"Total: {total}\nAverage: {average}\nGrade: {grade}"
    )

root = tk.Tk()
root.title("Grade Calculator")
root.geometry("350x350")

tk.Label(root, text="Student Grade Calculator",
        font=("Arial", 16)).pack(pady=10)

tk.Label(root, text="Subject 1").pack()
m1 = tk.Entry(root)
m1.pack()

tk.Label(root, text="Subject 2").pack()
m2 = tk.Entry(root)
m2.pack()

tk.Label(root, text="Subject 3").pack()
m3 = tk.Entry(root)
m3.pack()

tk.Button(root, text="Calculate",
        command=calculate).pack(pady=15)

result = tk.Label(root, text="")
result.pack()

root.mainloop()
```

Sample Output

Student Grade Calculator

Subject 1: 90
Subject 2: 80
Subject 3: 70

[Calculate]

Total: 240
Average: 80.0
Grade: B

4. Tkinter Login Form

Program Code

```
import tkinter as tk

def login():
    username = user.get()
    password = pwd.get()

    if username == "admin" and password == "1234":
        result.config(text="Login Successful")
    else:
        result.config(text="Invalid Username or Password")

root = tk.Tk()
root.title("Login Form")
root.geometry("350x300")

tk.Label(root, text="LOGIN FORM",
         font=("Arial", 18)).pack(pady=20)

tk.Label(root, text="Username").pack()
user = tk.Entry(root)
user.pack()

tk.Label(root, text="Password").pack()
pwd = tk.Entry(root, show="*")
pwd.pack()

tk.Button(root, text="Login",
          command=login).pack(pady=20)

result = tk.Label(root, text="")
result.pack()

root.mainloop()
```

Sample Output

LOGIN FORM

Username: admin

Password: ****

[Login]

Login Successful